

# **Quiz 8**

**STAT 109: Introductory Biostatistics**

# Quiz 8

## Quiz 8 Practice Problems

These practice problems cover **Lab 7** and **Lectures 18-21**:

- confidence intervals for one mean and one proportion,
  - when to use **FPC**,
  - checking interval conditions,
  - **convenience** vs **simple random sampling (SRS)**,
  - sampling frame and bias,
  - interpreting confidence intervals in context.
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## Summary box: interval forms, SE, and conditions

**i** Quick reference for Quiz 8

### 1) One mean, no FPC (infinite population / tiny sampling fraction)

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

- **Statistic:**  $\bar{x}$
- **SE:**  $s/\sqrt{n}$
- **Conditions:** random sample (or SRS), and population roughly normal or  $n$  moderately large

### 2) One mean, with FPC (finite population, sampling without replacement)

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

- Use when population is finite and  $n/N$  is not small.

### 3) One proportion, no FPC (large-sample normal / Wald form)

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- **Statistic:**  $\hat{p} = x/n$
- **SE:**  $\sqrt{\hat{p}(1-\hat{p})/n}$
- **Conditions:** SRS/random sample, independent outcomes, and large counts ( $n\hat{p} \geq 10$ ,  $n(1-\hat{p}) \geq 10$ )

### 4) One proportion, with FPC

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \sqrt{\frac{N-n}{N-1}}$$

- Use when sampling without replacement from known finite  $N$  and  $n/N$  is not small.

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## Instructions

- Work **without notes** first, then check your work.
- Show setup and reasoning, not only final numbers.
- Unless a problem says otherwise, use **95% confidence**.
- For interval problems, **check conditions using the summary table** before interpreting.

## CI interpretation template (use this wording pattern)

I am 95% confident that the [parameter] in the entire population is between [L] and [U]. In this context, confidence means that if data from many samples were collected the same way I did, then 95% of the intervals computed from these samples would contain the true value of the parameter.

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## Part A: Definitions and core ideas

### A1. Population

Define **population** in one or two clear sentences.

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### A2. Sample

Define **sample** in one or two clear sentences.

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### A3. Parameter

Define **parameter** in one or two clear sentences.

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### A4. Statistic

Define **statistic** in one or two clear sentences.

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### A5. Sampling frame

Define **sampling frame** in one or two clear sentences.

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### A6. Bias

Define **bias** in the context of sampling and estimation.

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### A7. Confidence level

Define **95% confidence level** in terms of repeated sampling and intervals.

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## A8. Confidence interval

Define **confidence interval** in one or two clear sentences.

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## Part B: Identify population, sample, parameter, statistic

For each scenario:

1. Identify the **population**.
  2. Identify the **sample**.
  3. Identify the **parameter**.
  4. Identify the **statistic** (and compute it if enough info is given).
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### B1.

A random sample of  $n = 120$  undergraduates is selected to estimate the proportion who work at least 10 hours/week. In the sample,  $x = 46$  say yes.

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### B2.

A district has  $N = 300$  classrooms. An SRS of  $n = 30$  classrooms is taken to estimate average number of chairs. The sample mean is  $\bar{x} = 29.8$  chairs.

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### B3.

A clinic wants the proportion of all current patients who are very satisfied. It samples  $n = 80$  patients this month, and 52 report “very satisfied.”

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### B4.

A researcher wants the mean number of minutes/day spent on course websites by all STAT 109 students. She samples 25 students and computes  $\bar{x} = 18.4$  minutes.

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## Part C: Convenience vs SRS, bias present or not

For each scenario:

1. Classify as **convenience** or **SRS**.
  2. Is bias likely present? (**Yes/No** + one sentence).
  3. State one sentence about whether results can reasonably generalize to the target population.
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### C1. Dining hall intercept

A student stands outside one dining hall from 12-1 p.m. and surveys the first 60 students who walk by about sleep habits.

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## C2. Registrar list sample

From a full registrar list of all undergraduates (ID 1 to  $N$ ), a random-number generator is used to select 120 unique IDs. Those students are surveyed.

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## C3. Club meeting sample

A researcher surveys 40 students attending the weekly pre-med club meeting to estimate the campus proportion planning to apply to medical school.

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## C4. SRS with missing frame coverage

An instructor creates a random sample from a list of students with declared majors only; undeclared students are not in the list.

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# Part D: Steps in SRS

List the SRS workflow in order using these three actions:

- list population units (or frame),
- generate random numbers,
- collect data.

Then apply those steps to the setting below.

## D1. Apply SRS steps

Goal: estimate mean weekly study hours for all STAT 109 students ( $N = 210$ ). Desired sample size:  $n = 30$ .

Write 3-4 bullets describing exactly how you would carry out the SRS.

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# Part E: Identify interval type, FPC, and conditions

For each scenario:

1. Is the parameter a **mean** or **proportion**?
  2. Use **FPC** or **no FPC**?
  3. Are the interval conditions reasonably met? (brief justification)
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## E1.

Sample of  $n = 220$  students;  $x = 82$  have a part-time job. The sample is SRS from a very large statewide population.

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## E2.

A district has  $N = 300$  classrooms. SRS without replacement gives  $n = 30$  classroom chair counts; you want a CI for average chairs/classroom.

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### E3.

From  $N = 80$  patients at a clinic,  $n = 50$  are sampled without replacement. Outcome is Yes/No (successful recovery), and you want a CI for the population success proportion.

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### E4.

SRS of  $n = 18$  students from a large campus. Outcome is whether the student is a biology major; observed  $\hat{p} = 2/18$ .

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## Part F: Interpret already-calculated intervals in context

For each item, write a correct interpretation sentence in context.

### F1. Proportion

A 95% CI for the proportion of all undergraduates who are first-generation college students is  $[0.22, 0.30]$ .

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### F2. Mean

A 95% CI for the mean number of minutes/day students spend on course websites is  $[41, 55]$  minutes.

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### F3. Finite population proportion

A school district sampled from a finite list of schools and computed a 95% CI (with FPC) for the proportion of schools older than 50 years:  $[0.14, 0.27]$ .

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### F4. Spot the incorrect interpretation

Choose the **best** interpretation of a 95% CI and explain briefly why:

- **A.** “There is a 95% probability that the true parameter is in this specific interval.”
  - **B.** “If we repeatedly took samples the same way and built intervals this way, about 95% of those intervals would contain the true parameter.”
  - **C.** “95% of individuals in the population fall between the interval endpoints.”
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## Part G: R output interpretation (read code and output)

For each problem, do **not** recompute by hand. Use the shown output.

1. Identify whether it is a CI for a **mean** or **proportion**.
2. Write the 95% CI from the output.
3. Interpret it in context in one sentence.

### Context for G1

A researcher wants to estimate the **proportion of CA high school students who report having a part-time job**.

```
# Example 1: proportion CI from a sample
n <- 220
x <- 80
prop.test(x, n, conf.level = 0.95, correct = FALSE)
```

1-sample proportions test without continuity correction

```
data: x out of n, null probability 0.5
X-squared = 16.364, df = 1, p-value = 5.228e-05
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.3029152 0.4290379
sample estimates:
      p
0.3636364
```

### G1. Questions for the output above

- Is this estimating a mean or a proportion?
  - What is the 95% interval (from the printed output)?
  - Write a context interpretation.
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### Context for G2

An instructor wants to estimate the **mean number of minutes per day students spend on the course website**.

```
# Example 2: mean CI from quantitative data
x <- c(12, 18, 25, 9, 31, 16, 20, 22, 14, 3, 19, 11, 24, 30, 15, 15, 21, 26, 13, 28, 10, 23, 29, 18, 33)
t.test(x, conf.level = 0.95)
```

One Sample t-test

```
data: x
t = 12.558, df = 24, p-value = 4.853e-12
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 16.21152 22.58848
sample estimates:
mean of x
 19.4
```

### G2. Questions for the output above

- Is this estimating a mean or a proportion?
  - What is the 95% interval (from the printed output)?
  - Write a context interpretation.
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*End of Quiz 8 practice problems.*